

1.1 Purpose of the Document

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements of the Quotewise System. It serves as a reference for developers, designers, and stakeholders to ensure the system meets user needs and project objectives.

1.2 Product Scope

The Quotewise system is designed to connect clients with construction companies by allowing clients to post construction requests and receive quotations from multiple companies.

Features INCLUDED:

User registration and login (client & company)

Posting construction project requests

Viewing available project requests (companies)

Submitting quotations (companies)

Viewing and comparing quotations (clients)

Selecting a preferred quotation

Features NOT INCLUDED:

Online payments

Contract signing functionality

Real-time chat system

AI-based contractor recommendations (future improvement)

1.3 Target Engineering Domain

This project falls under:

Civil Engineering

1.4 Problem Statement

Currently, clients seeking construction services rely on manual processes such as phone calls, physical visits, or word-of-mouth to obtain quotations.

Existing systems suffer from:

Lack of centralized platforms

Time-consuming quotation processes

Limited access to multiple contractors

The proposed system aims to solve these challenges by providing a centralized digital platform where clients can easily request services and compare quotations from multiple construction companies.

2. SYSTEM OVERVIEW

2.1 System Description

The QW system operates as an online platform where clients post construction project descriptions. Construction companies can view these requests and decide whether to submit quotations.

The system allows users to:

Post construction needs

Submit and receive quotations

Compare available offers

It collects, processes, and stores data to support decision-making.

2.2 Target Users

User Type

Description

Technical Level

Client

Individual seeking construction services

Basic

Construction Company

Business providing construction service (Intermediate)

System Admin (optional)

Maintains system operations

Advanced

2.3 Technology Overview

The system will be developed using the following technologies:

Frontend:

React Native (for mobile application development)

Backend:

Node.js (for server-side logic and API handling)

Database / Cloud Services:

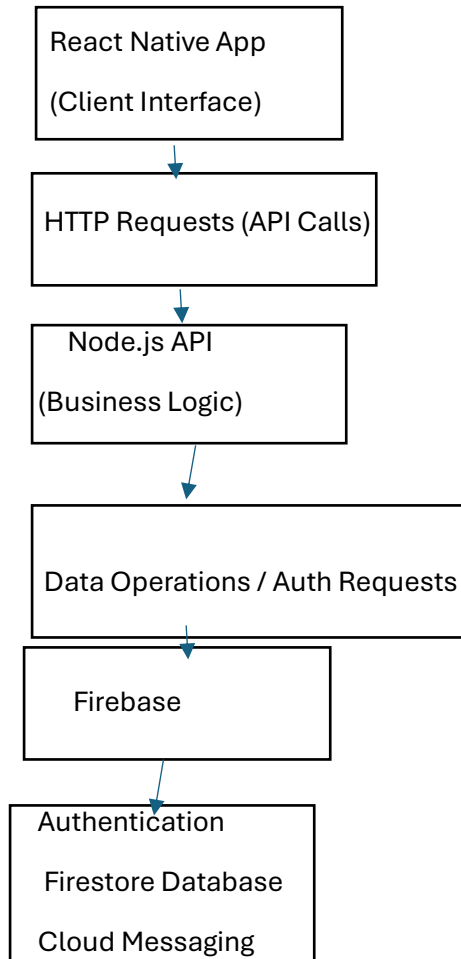
Firebase

Firebase Authentication (user login & signup)

Fire store Database (data storage)

Firebase Cloud Messaging (notifications)

- **System Architecture Diagram**



2.4 Data Model

Users:

User ID

Name

Role (Client / Company)

Email

Password

Contact Information

Projects:

Project ID

Client ID

Description

Budget (optional)

Date Posted

Status

Quotations:

Quotation ID

Project ID

Company ID

Price

Timeline

Description

Status

2.5 Assumptions and Constraints

Assumptions:

Users have smartphones or computers

Internet access is available

Users provide accurate information

Constraints:

Dependent on internet connectivity

Limited initial budget for development

System scalability may be limited in early versions

3. FUNCTIONAL REQUIREMENTS

FR-01: Users shall be able to register and log in.

Priority: High

FR-02: Clients shall be able to post construction project descriptions.

Priority: High

FR-03: Construction companies shall be able to view posted projects.

Priority: High

FR-04: Construction companies shall be able to submit quotations.

Priority: High

FR-05: Clients shall be able to view all submitted quotations.

Priority: High

FR-06: Clients shall be able to select a preferred quotation.

Priority: High

FR-07: The system shall store all user and project data securely.

Priority: High

FR-08: The system shall notify users when a quotation is received.

Priority: Medium

FR-09: Users shall be able to edit or delete their posts.

Priority: Medium

4. NON-FUNCTIONAL REQUIREMENTS

Performance:

The system shall respond within 2–3 seconds under normal load

Security:

Authentication required for all users

Password encryption must be implemented

Role-based access control

Usability:

Simple and intuitive interface

Mobile-friendly design

Compatibility:

Runs on web browsers (Chrome, Edge, Firefox)

Accessible on mobile and desktop

Reliability:

System uptime of at least 95%

Automatic data backup

5. USE CASE DESCRIPTION

Use Case 1: Post Construction Request

Actor: Client

Description:

The client logs into the system and submits a construction project description including requirements and optional budget.

Outcome:

The request is stored and visible to construction companies.

Use Case 2: Submit Quotation

Actor: Construction Company

Description:

The company views available projects and submits a quotation including price and timeline.

Outcome:

The quotation is stored and visible to the client.

Use Case 3: Select Quotation

Actor: Client

Description:

The client reviews multiple quotations and selects the preferred one.

Outcome:

The selected company is marked as chosen, and the project status is updated.

6. CONCLUSION

The QuoteWise (QW) is expected to improve efficiency in the construction service process by digitizing how clients and contractors interact.